Cross-Sell Targeting Model - Brief for Q4 Campaign

From: Head of CRM & Cross-Sell (Marketing)

To: Data Science team

Date: 1 October 2026

Decision needed by: 15 October 2026 Q4 campaign planning meeting

Background

Our campaigns offer an additional product (credit card, savings, personal loan or insurance) to existing customers. So far we have contacted customers at random and converted roughly 1 in 9. We want to stop spending budget on customers who are very unlikely to buy.

The Data Warehouse team has shared a snapshot of 20,000 customers as of 30 June 2026, together with whether each customer opened an additional product in the following 90 days (July - September). File: input 1 - customers.csv, definitions in input 2 - data dictionary.xlsx.

Before we commit the Q4 budget we want evidence: if we had used a model on 30 June, how many more customers would we have converted than with random selection?

What we need

1. A model that ranks customers by their likelihood to buy another product in the next 90 days, packaged so we can score the 30 September snapshot for Q4.

2. A back-test target list: the 4,000 customers the model would have selected on 30 June, scored only with models that did not train on those customers, with expected and actual conversions and ROI compared to random selection.

3. The main drivers of purchase in plain business language, so the creative team can tailor messaging.

4. A short (3-slide) summary for the campaign planning meeting, using the team template (input 4 - summary deck template.pptx).

Business constraints

- Budget: 4,000 contacts per campaign. Cost per contact (incl. incentive): $4.
- Value: average 12-month margin per additional product sold: $150.
- Compliance: only customers with marketing_opt_in = 1 may be contacted.
- Compliance: gender must NOT be used as a model input (fair-treatment policy).
- Customer care rule: exclude customers with 2 or more complaints in the last 12 months from the target list.
- The model must only use information that is available on the snapshot date - it will be re-run on future snapshots before each campaign.

Acceptance criteria

- Performance on a hold-out set the model never saw: ROC-AUC of at least 0.78.

- Beats a simple logistic-regression baseline on PR-AUC (or the reason is explained).
- Top 20% of scored customers convert at least 2.5x the average rate (lift >= 2.5).
- No leakage: no field that is only populated after the purchase is used.
- Back-test list honours every compliance and customer-care rule above, has exactly 4,000 rows, and its actual conversion rate is at least 2x the eligible-customer average.
- Results can be reproduced by re-running the pipeline notebooks end to end.